

Features

- Linear current limitation
- Thermal shutdown
- Short circuit protection
- Integrated clamp
- Low current drawn from input pin
- Diagnostic feedback through input pin
- ESD protection
- Direct access to the gate of the Power MOSFET (analog driving)
- Compatible with standard Power MOSFET
- Automotive qualified (AEC-Q100 Grade 1)



Description

The HELS120D device is made up of two monolithic chips housed in a standard ESOP-8 package, is intended for replacement of standard Power MOSFETs in up to 50 kHz DC applications.

Built-in thermal shutdown, linear current limitation and overvoltage clamp protect the chip in harsh environments.

Fault feedback can be detected by monitoring voltage at the input pin.

Type	R _{DS(on)}	I _{lim}	V _{clamp}
HELS120D-ESOP8	120mΩ	2.3A	40V

Part number	Package	Marking	Packing
HELS120DRBQC	ESOP-8	HELS120D	3000 / Tape & Reel

*Not sure if you got the latest datasheet or not. Please kindly contact the sales of KKChips to confirm. Thanks.

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1 Block diagram and pin description

1.1 Block diagram

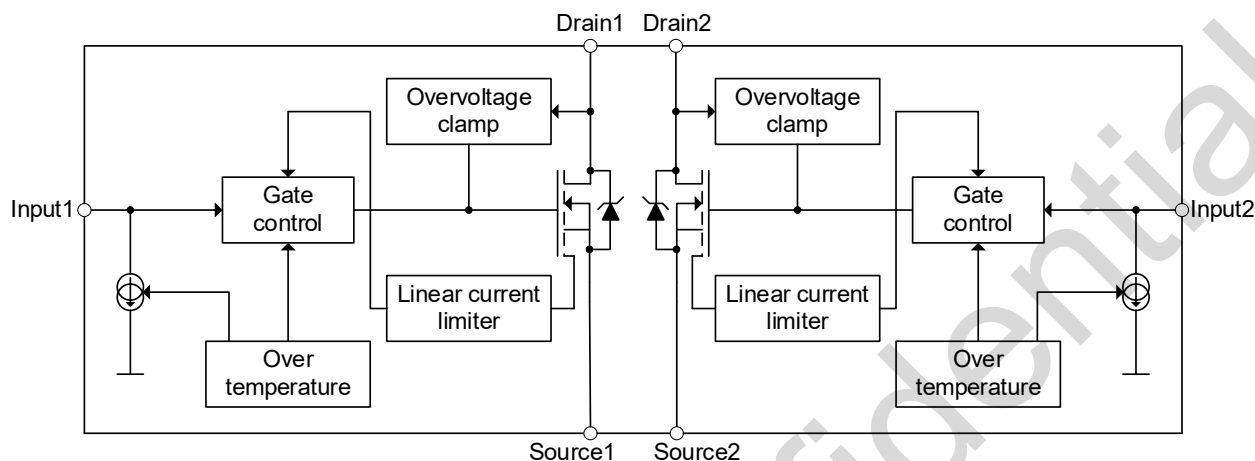


Figure 1-1 Block diagram

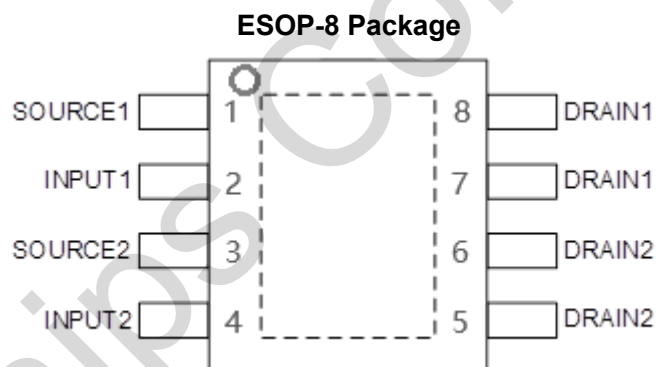


Figure 1-2 Configuration diagram (top view)

2 Electrical specifications

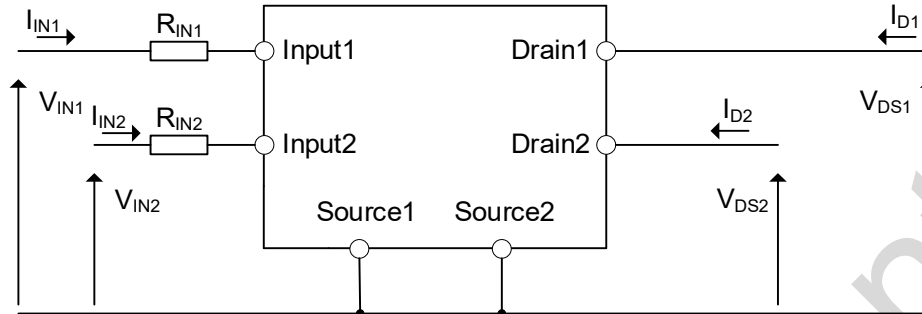


Figure 2-1 Current and voltage conventions

2.1 Absolute maximum ratings

Stressing the device above the rating listed in the “Absolute maximum ratings” table may cause permanent damage to the device. These are stress ratings only and operation of the device at these or any other conditions above those indicated in the operating sections of this specification is not implied. Exposure to Absolute maximum rating conditions for extended periods may affect device reliability.

Table 2-1 Absolute maximum ratings

Symbol	Parameter	Value	Unit
V_{DSn}	Drain-Source Voltage ($V_{INn} = 0\text{ V}$)	Internally clamped	V
V_{INn}	Input voltage	Internally clamped	V
I_{INn}	Input current	+/- 20	mA
$R_{IN\ MINn}$	Minimum input series impedance	220	Ω
I_{Dn}	Drain current	Internally limited	A
I_{Rn}	Reverse DC output current	-5.5	A
V_{ESD1}	Electrostatic discharge ($R = 1.5\text{ K}\Omega$, $C = 100\text{ pF}$)	4000	V
P_{tot}	Total dissipation at $T_c = 25^\circ\text{C}$	4	W
T_j	Operating junction temperature	Internally limited	$^\circ\text{C}$
T_c	Case operating temperature	Internally limited	$^\circ\text{C}$
T_{stg}	Storage temperature	-55 to 150	$^\circ\text{C}$

2.2 Thermal data

Table 2-2 Thermal data

Symbol	Parameter	Max Value	Unit
$R_{thj-case}$	Thermal resistance junction-case max	18	°C/W
$R_{thj-amb}$	Thermal resistance junction-ambient	80 ⁽¹⁾	°C/W

1. When mounted on a standard single-sided FR4 board with 50mm² of Cu (at least 35 μm thick) connected to all DRAIN pins of the relative channel

2.3 Electrical characteristics

Values specified in this section are for $-40^{\circ}\text{C} < T_j < 150^{\circ}\text{C}$, unless otherwise stated.

Table 2-3 Off

Symbol	Parameter	Condition	Min	Typ	Max	Unit
V_{CLAMP}	Drain-source clamp voltage	$V_{IN} = 0\text{ V}; I_D = 1.5\text{ A}$	40	44.5	55	V
V_{CLTH}	Drain-source clamp threshold voltage	$V_{IN} = 0\text{ V}; I_D = 2\text{ mA}$	36	44	-	V
V_{INTH}	Input threshold voltage	$V_{DS} = V_{IN}; I_D = 1\text{ mA}$	0.5	1.6	2.5	V
I_{ISS}	Supply current from input pin	$V_{DS} = 0\text{ V}; V_{IN} = 5\text{ V}$	-	87	150	μA
V_{INCL}	Input-source clamp voltage	$I_{IN} = 1\text{ mA}$	6	6.6	7	V
		$I_{IN} = -1\text{ mA}$	-1.0	-	-0.3	V
I_{DSS}	Zero input voltage drain current ($V_{IN} = 0\text{ V}$)	$V_{DS} = 13\text{ V}; V_{IN} = 0\text{ V}; T_j = 25^{\circ}\text{C}$	-	-	10	μA
		$V_{DS} = 25\text{ V}; V_{IN} = 0\text{ V}$	-	-	10	μA

Table 2-4 On

Symbol	Parameter	Condition	Min	Typ	Max	Unit
$R_{DS(on)}$	Static drain-source on resistance	$V_{IN} = 5\text{ V}; I_D = 1.5\text{ A}; T_J = 25^\circ\text{C}$	-	-	120	m Ω
		$V_{IN} = 5\text{ V}; I_D = 1.5\text{ A}$	-	-	240	m Ω

Table 2-5 Dynamic ($T_J = 25^\circ\text{C}$, unless otherwise specified)

Symbol	Parameter	Condition	Min	Typ	Max	Unit
G_{fs}	Forward transconductance	$V_{DD} = 13\text{ V}; I_D = 1.5\text{ A}$	-	5.0	-	S
C_{OSS}	Output capacitance	$V_{DS} = 13\text{ V}; f = 1\text{ MHz}; V_{IN} = 0\text{ V}$	-	150	-	pF

Table 2-6 Switching ($T_J = 25^\circ\text{C}$, unless otherwise specified)

Symbol	Parameter	Condition	Min	Typ	Max	Unit
$t_{d(on)}$	Turn-on delay time	$V_{DD} = 15\text{ V}; I_D = 1.5\text{ A};$ $V_{gen} = 5\text{ V}; R_{gen} = R_{IN\text{ MIN}} = 220\text{ }\Omega$ (see Figure 2-2)	-	100	300	ns
t_r	Rise time		-	400	900	ns
$t_{d(off)}$	Turn-off delay time		-	500	1350	ns
t_f	Fall time		-	200	750	ns
$t_{d(on)}$	Turn-on delay time	$V_{DD} = 15\text{ V}; I_D = 1.5\text{ A};$ $V_{gen} = 5\text{ V}; R_{gen} = 2.2\text{ K}\Omega$ (see Figure 2-2)	-	0.4	1.35	μs
t_r	Rise time		-	0.65	3	μs
$t_{d(off)}$	Turn-off delay time		-	2.6	8	μs
t_f	Fall time		-	1.4	6	μs
$(di/dt)_{on}$	Turn-on current slope	$V_{DD} = 15\text{ V}; I_D = 1.5\text{ A}; V_{gen} = 5\text{ V};$ $R_{gen} = R_{IN\text{ MIN}} = 220\Omega$	-	3	-	A/ μS
Q_i	Total input charge	$V_{DD} = 12\text{ V}; I_D = 1.5\text{ A}; V_{IN} = 5\text{ V};$ $I_{gen} = 2.13\text{ mA}$ (see Figure 2-5)	-	10	-	nC

Table 2-7 Source drain diode ($T_j = 25^\circ\text{C}$, unless otherwise specified)

Symbol	Parameter	Condition	Min	Typ	Max	Unit
$V_{SD}^{(1)}$	Forward on voltage	$I_{SD} = 1.5\text{ A}$; $V_{IN} = 0\text{ V}$	-	0.82	-	V
t_{rr}	Reverse recovery time	$I_{SD} = 1.5\text{ A}$; $dI/dt = 12\text{ A}/\mu\text{s}$; $V_{DD} = 30\text{ V}$; $L = 200\mu\text{H}$ (see Figure 2-3)	-	107	-	ns
Q_{rr}	Reverse recovery charge		-	37	-	μC
I_{RRM}	Reverse recovery current		-	0.7	-	A

Table 2-8 Protections ($-40^\circ\text{C} < T_j < 150^\circ\text{C}$, unless otherwise specified)

Symbol	Parameter	Condition	Min	Typ	Max	Unit
I_{lim}	Current limitation range	$V_{IN} = 5\text{ V}$; $V_{DS} = 13\text{ V}$;	2.3	2.67	4	A
I_{lim_th}	Current limitation trigger threshold	$V_{IN} = 5\text{ V}$; $V_{DS} = 13\text{ V}$; Guarantee by design	2	5	9	A
		$V_{IN} = 5\text{ V}$; $V_{DS} = 13\text{ V}$; $T_j = 25^\circ\text{C}$; Guarantee by design	3.5	6	6.5	A
		$V_{IN} = 5\text{ V}$; $V_{DS} = 13\text{ V}$; $T_j = 125^\circ\text{C}$; Guarantee by design	2	-	4	A
t_{dlim}	Step response current limit	$V_{IN} = 5\text{ V}$; $V_{DS} = 13\text{ V}$	-	10	-	μs
T_{jsh}	Overtemperature shutdown	-	150	175	200	$^\circ\text{C}$
T_{jrs}	Overtemperature reset	-	135	-	-	$^\circ\text{C}$
I_{gf}	Fault sink current	$V_{IN} = 5\text{ V}$; $V_{DS} = 13\text{ V}$; $T_j = T_{jsh}$	10	15	20	mA
E_{as}	Single pulse avalanche energy	Starting $T_j = 25^\circ\text{C}$; $V_{DD} = 24\text{ V}$; $V_{IN} = 5\text{ V}$; $R_{gen} = R_{IN\text{ MIN}}$ $= 220\Omega$; $L = 24\text{ mH}$ (see Figure 2-4 and Figure 2-6)	100	-	-	mJ

Protection features

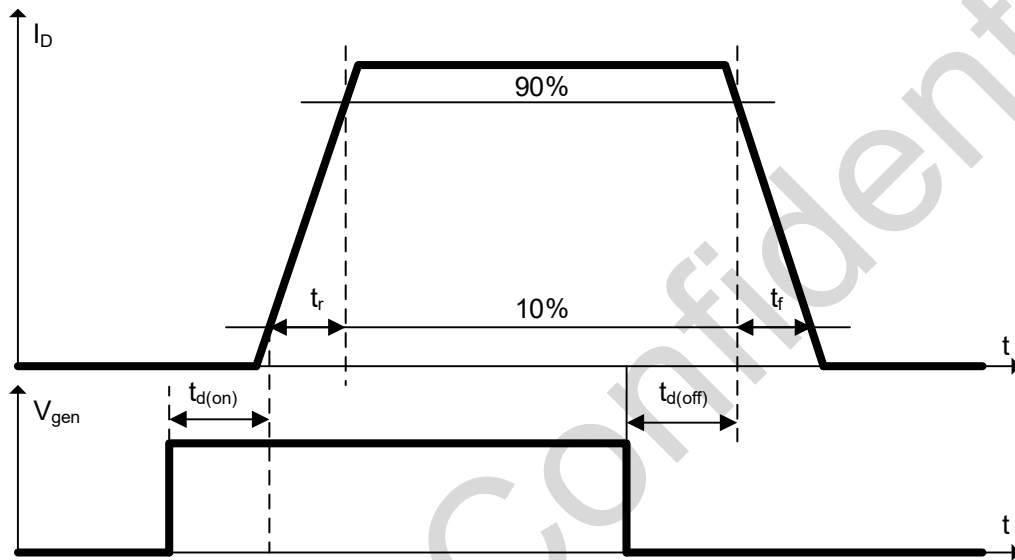
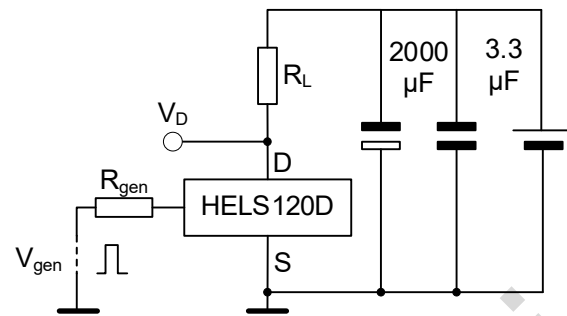


Figure 2-2 Switching time test circuit for resistive load

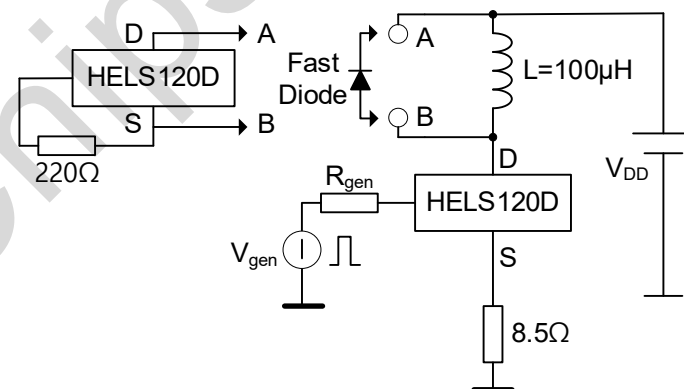


Figure 2-3 Test circuit for diode recovery times

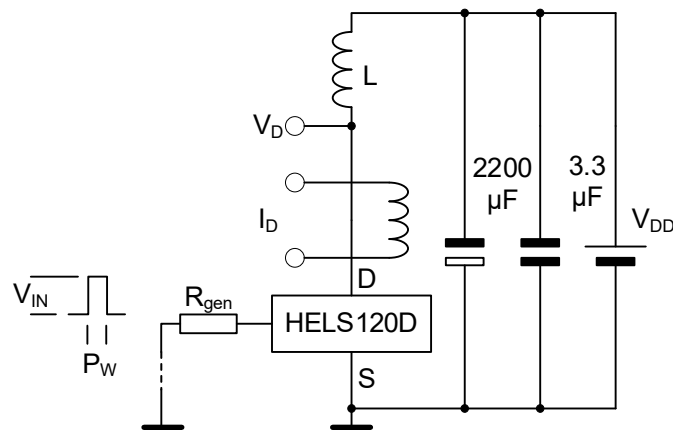


Figure 2-4 Unclamped inductive load test circuits

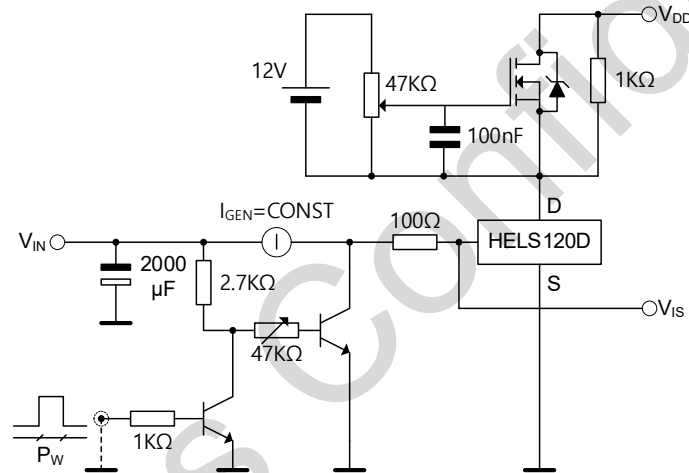


Figure 2-5 Input charge test circuit

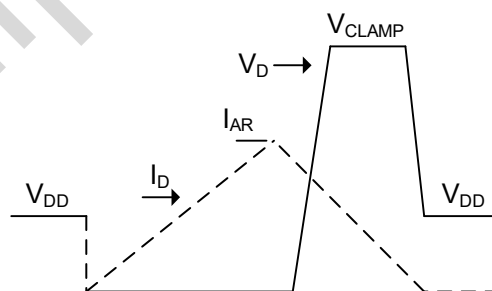


Figure 2-6 Unclamped inductive waveforms

2.4 Electrical characteristics curves

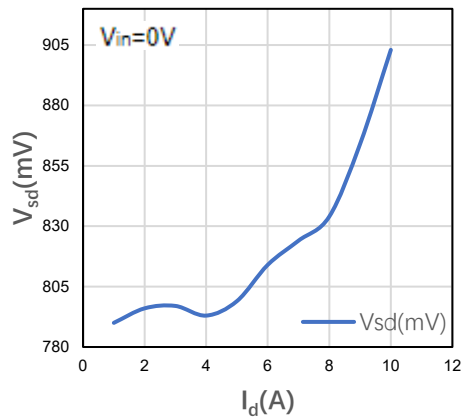


Figure 2-7 Source-drain diode forward characteristics

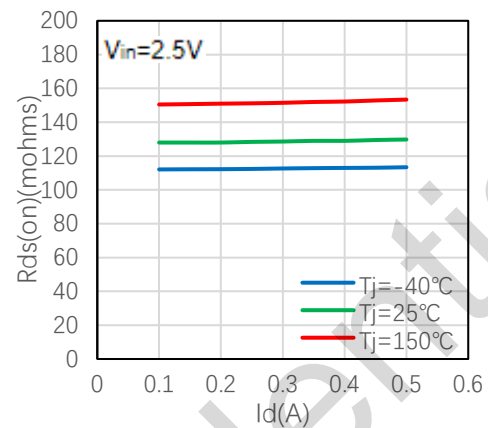


Figure 2-8 Static drain-source on resistance

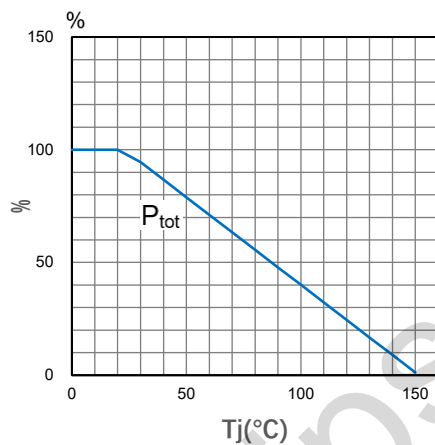


Figure 2-9 Derating curve

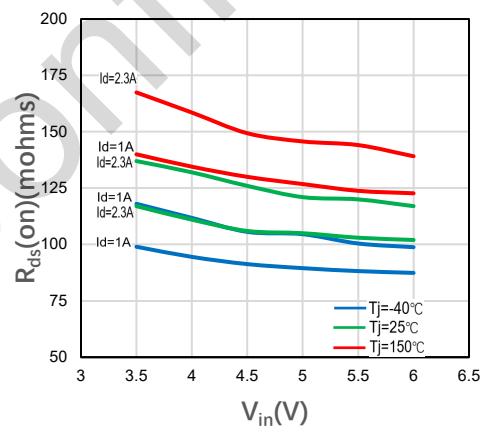


Figure 2-10 Static drain-source on resistance vs input voltage (part 1)

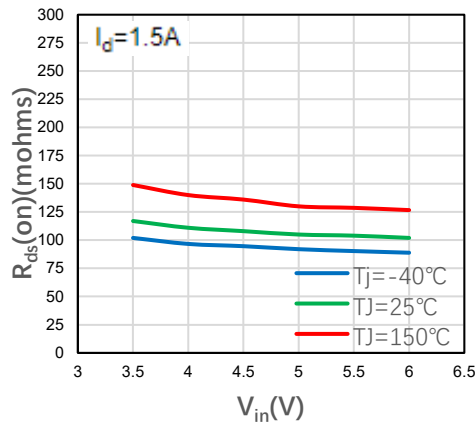


Figure 2-11 Static drain-source on resistance vs input voltage (part 2)

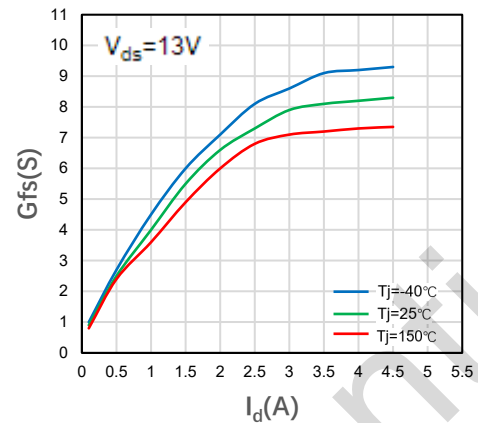


Figure 2-12 Transconductance

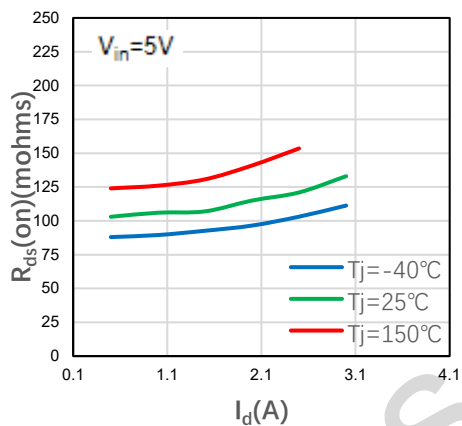


Figure 2-13 Static drain-source on resistance vs Id

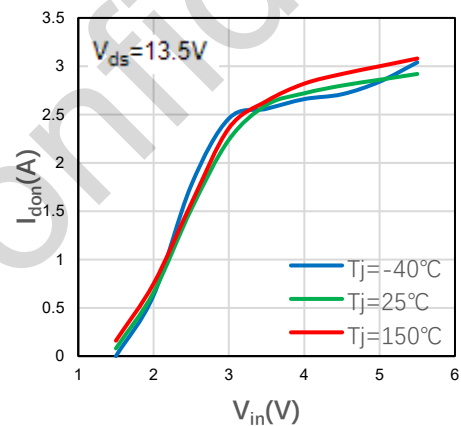


Figure 2-14 Transfer characteristics

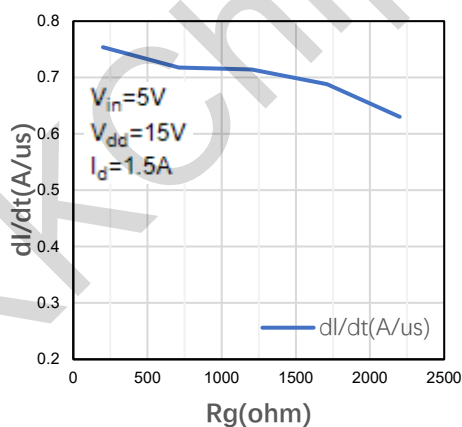


Figure 2-15 Turn-on current slope (part 1)

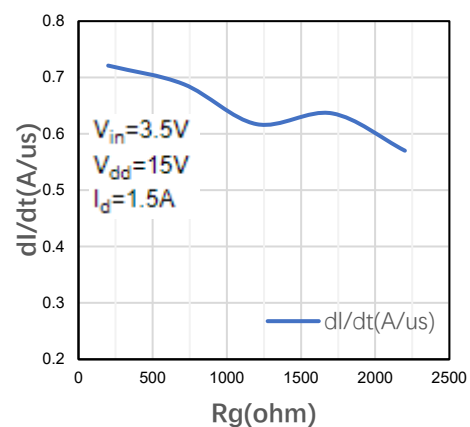


Figure 2-16 Turn-on current slope (part 2)

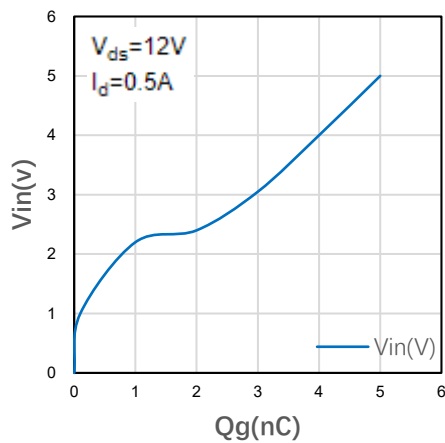


Figure 2-17 Input voltage vs input charge

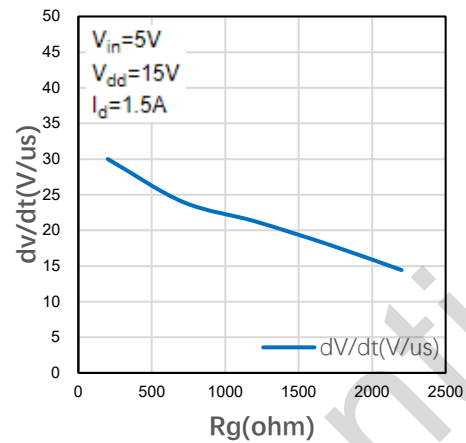


Figure 2-18 Turn-off drain-source voltage slope (part 1)

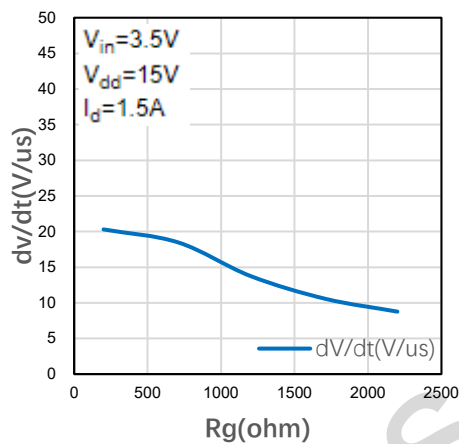


Figure 2-19 Turn-off drain-source voltage slope (part 2)

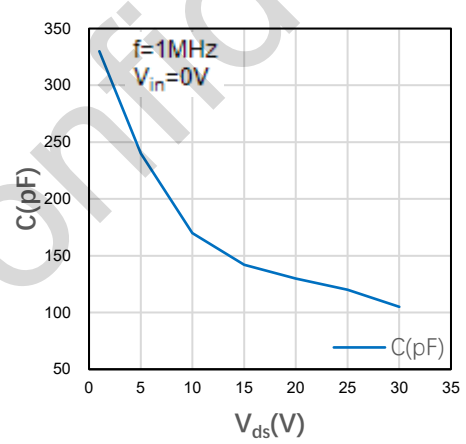


Figure 2-20 Capacitance variations

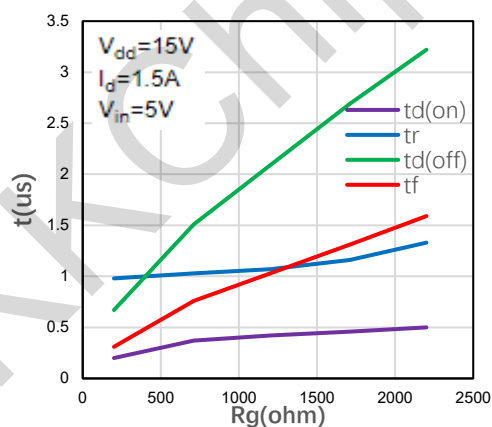


Figure 2-21 Switching time resistive load (part 1)

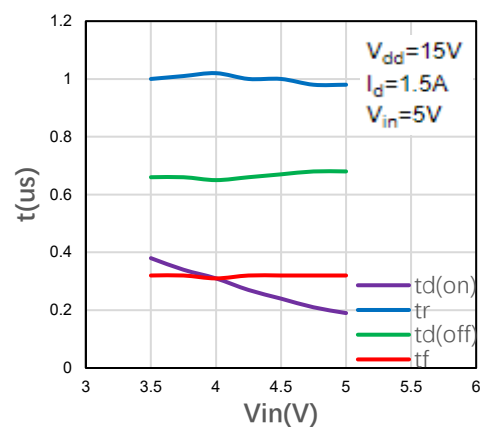


Figure 2-22 Switching time resistive load (part 2)

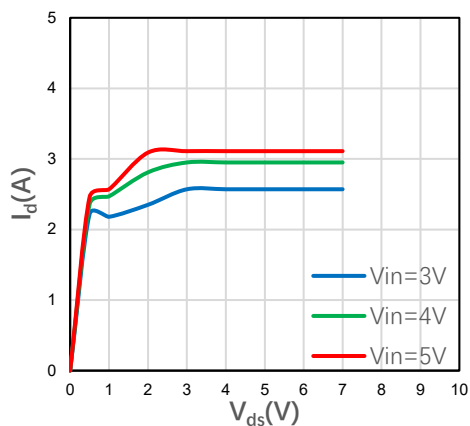


Figure 2-23 Output characteristics

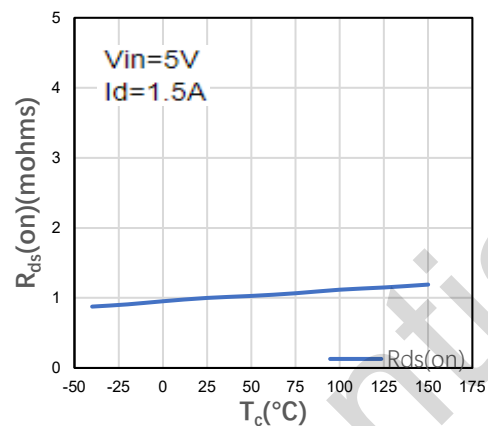


Figure 2-24 Normalized on resistance vs temperature

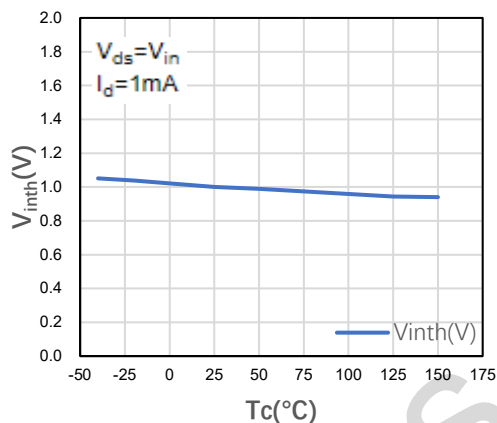


Figure 2-25 Normalized input threshold voltage vs temperature

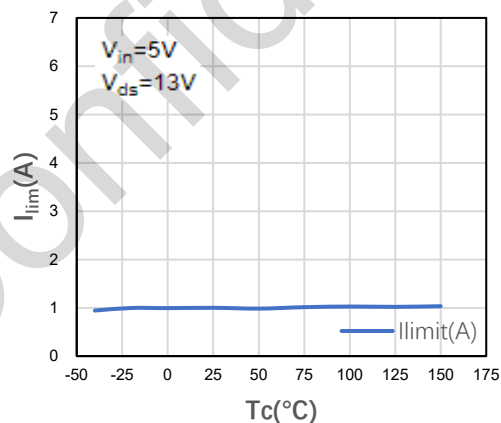


Figure 2-26 Normalized current limit vs junction temperature

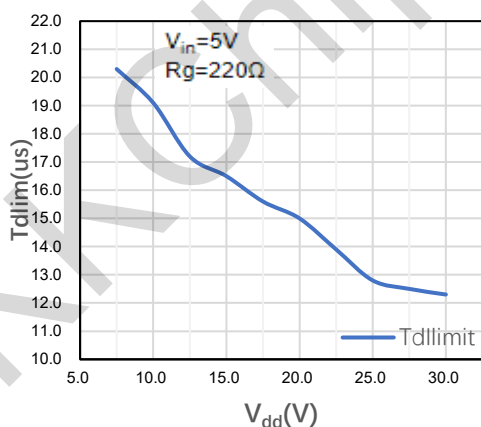


Figure 2-27 Step response current limit

3 Protection features

During normal operation, the INPUT pin is electrically connected to the gate of the internal power MOSFET through a low impedance path.

The device then behaves like a standard power MOSFET and can be used as a switch from DC up to 50 KHz. The only difference from the user's standpoint is that a DC current flows into the INPUT pin in order to supply the internal circuitry.

The following sections describe the device features.

3.1 Overvoltage clamp protection

Internally set at 45 V, along with the rugged avalanche characteristics of the Power MOSFET stage give this device unrivalled ruggedness and energy handling capability. This feature is mainly important when driving inductive loads.

3.2 Linear current limiter circuit

Limits the drain current I_D to I_{lim} whatever the INPUT pin voltages. When the current limiter is active, the device operates in the linear region, so power dissipation may exceed the capability of the heatsink. Both case and junction temperatures increase, and if this phase lasts long enough, junction temperature may reach the overtemperature threshold T_{jsh} .

3.3 Overtemperature and short circuit protection

These are based on sensing the chip temperature and are not dependent on the input voltage. The location of the sensing element on the chip in the power stage area ensures fast, accurate detection of the junction temperature. Overtemperature cutout occurs in the range 150 to 190 °C, a typical value being 170 °C. The device is automatically restarted when the chip temperature falls of about 15 °C below shutdown temperature.

3.4 Status feedback

In the case of an overtemperature fault condition ($T_j > T_{jsh}$), the device tries to sink a diagnostic current I_{gf} through the INPUT pin in order to indicate fault condition. If driven from a low impedance source, this current may be used in order to warn the control circuit of a device shutdown. If the drive impedance is high enough so that the INPUT pin driver is not able to supply the current I_{gf} , the INPUT pin falls to 0 V. This however not affects the device operation: no requirement is put on the current capability of the INPUT pin driver except to be able to supply the normal operation drive current I_{ISS} .

Additional features of this device are ESD protection according to the Human Body model and the ability to be driven from a TTL Logic circuit.

4 Package and packing information

4.1 ESOP-8 mechanical data

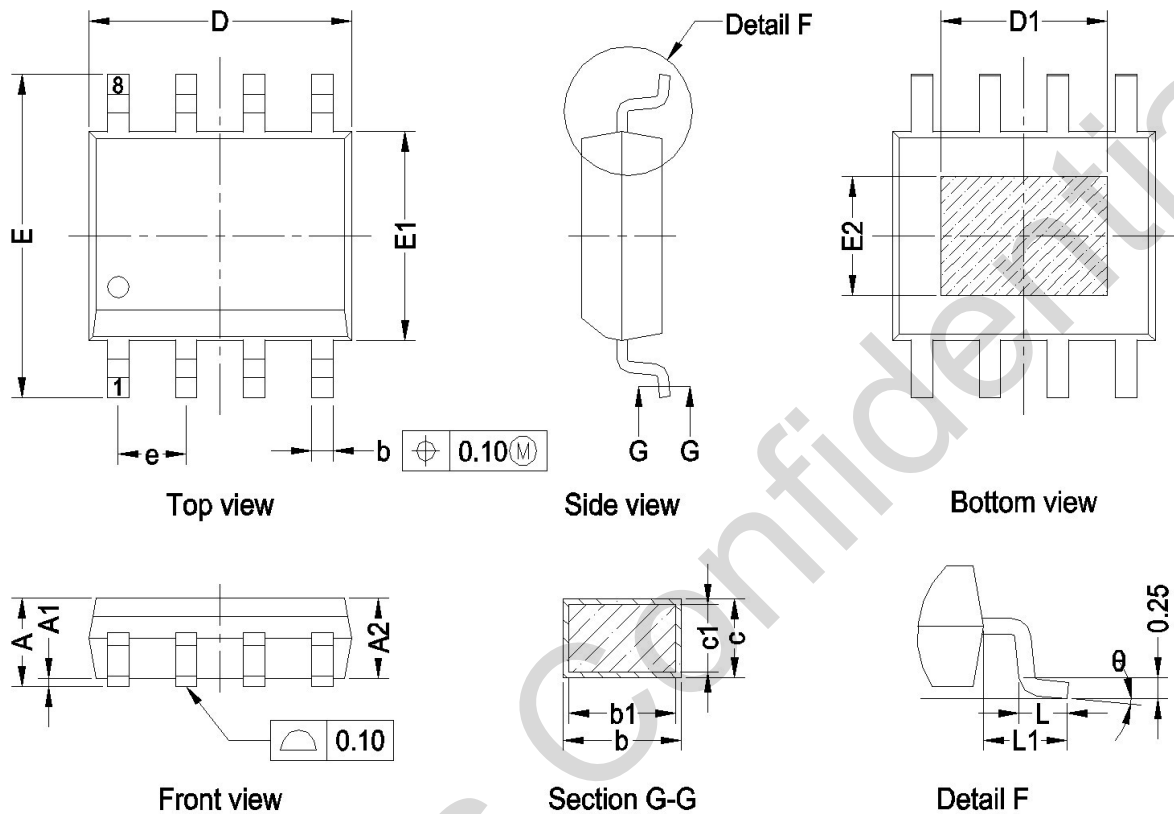


Figure 4-1 ESOP-8 package dimension

Table 4-1 ESOP-8 mechanical data

Dim.	mm			Description
	Min.	Typ.	Max.	
A	-	-	1.70	Total Thickness
A1	0.05	0.10	0.15	Stand Off
A2	1.30	1.45	1.55	Mold Thickness
b	0.39	-	0.51	Lead Width(Plating)
b1	0.36	0.41	0.46	Lead Width
c	0.17	-	0.25	L/F Thickness(Plating)

Package and packing information

c1	0.203 REF			L/F Thickness
D	4.70	4.90	5.10	Body Size(Length)
D1	3.00	-	3.25	EP Size(length)
E	5.80	6.00	6.20	Lead Tip to Tip
E1	3.80	3.90	4.00	Body Size(Width)
E2	2.10	-	2.35	EP Size(Width)
e	1.27 BSC			Lead Pitch
L	0.40	0.6	0.80	Foot Length
L1	1.00 REF			Foot Print
Θ	0°	-	8°	Angle

5 Recommended furnace temperature curve

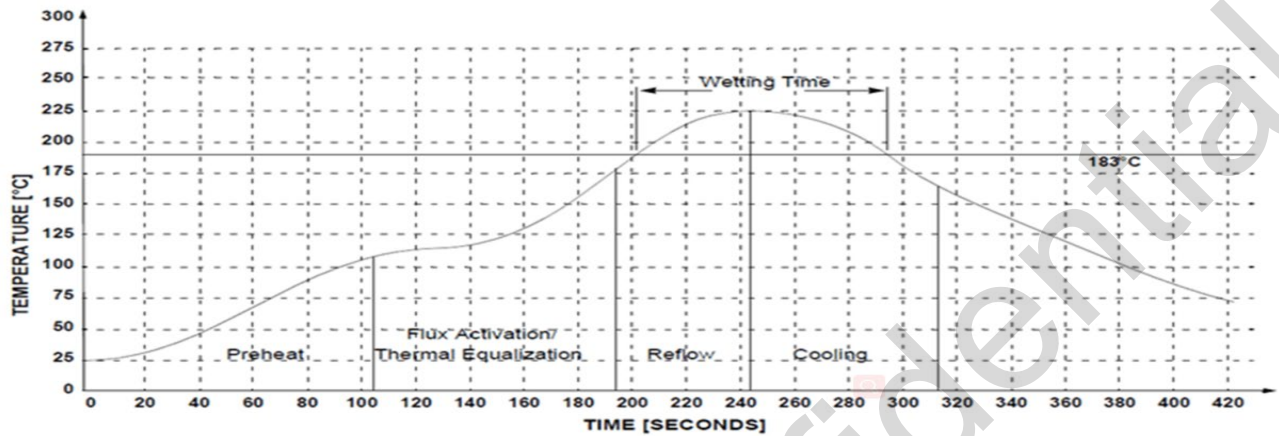


Figure 5-1 Recommended furnace temperature curve

6 Recommended pad size

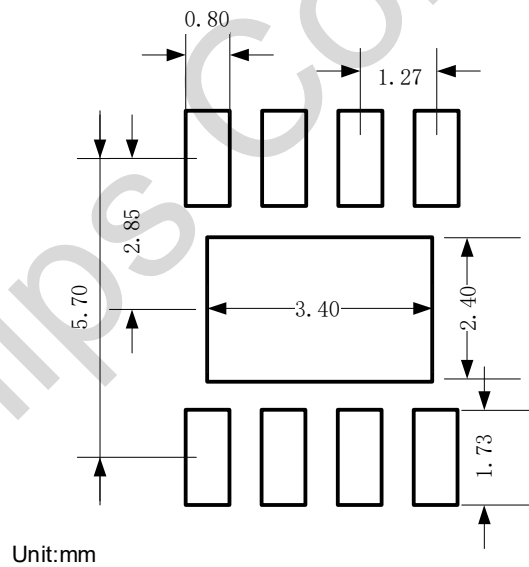


Figure 6-1 Recommended pad size

7 Customer Service

KKchips

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Revision history

Date	Revision	Changes
2022/10/11	1.1	- Updated "Note *" in the Section 2 Electrical specifications; - Updated Figure 2-1 Current and voltage conventions.
2023/03/13	1.2	- Update typography; - Updated Part number; - Update charts; - Remove SO-8 related data.
2023/09/12	1.3	1.Format updated to 1.3
2024/02/05	1.4	1. Format updated to 1.4.
2024/04/22	1.5	- Package information update - Format updated to 1.5.
2024/07/04	1.6	1.Changed the format of the curve in Section 2.4
2024/09/03	1.7	1. Updated the company's new address
2024/12/18	1.8	- Format updated to 1.8. - Modify tdon_220 parameters from 265ns to 100ns. - Modified tr_220 parameters from 1630ns to 400ns, and the maximum value from 1800ns to 900ns. - Modified tdoff_220 parameter from 670ns to 500ns, maximum value from 800ns to 1350ns - Modified tf_220 parameter from 340ns to 200ns, maximum value from 400ns to 750ns - Modified tdon_2200 parameters from 0.6us to 0.4us, maximum value from 0.8us to 1.35us - Revised tr_2200 parameters from 1.8us to 0.65us, maximum value from 2us to 3us - Modified tdoff_2200 parameters from 3.2us to 2.6us, maximum value from 3.5us to 8us. - Revised tf_2200 parameters from 1.6us to 1.4us, maximum value from 2us to 6us - Revised dI/dt parameter from 0.7A/us to 3A/us. - Revised the description in 3 Protection features.
2025/1/23	1.9	1. Revision function description.
2025/4/24	2.0	1. The maximum value of Vincl has been changed from 8V to 7V.

		2. Corrected data errors in V_{SD} , from 0.96V to 0.82V
2026/6/3	2.1	1. The I limit range is revised from 2.3A~3.3A to 2.3A~4A. 2. Revision Format.

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